HLAbAssist Complete Guide: Exporting and Analyzing SAB Data

Purpose: End-to-end instructions for exporting Single Antigen Bead (SAB) data from your analysis software and uploading it to HLAbAssist (hlabassist.app) for antibody identification, DSA identification, and virtual crossmatch prediction.

Audience: HLA laboratory personnel, transplant immunologists, and researchers using One Lambda Fusion (LABScreen) or Werfen MATCH IT! Antibody (LIFECODES LSA) software.

Time required: 5–15 minutes for first-time export template setup; 2–5 minutes per patient after that.

Overview

This guide is organized in two parts:

Part 1: Export from your SAB software

Section A: One Lambda Fusion

- A1: Create the Export Template (One-Time Setup)
- A2: Export Your SAB Data (multi-patient/batch)
- A3: Exporting a Single Patient from a Multi-Patient Session
- A4: Troubleshooting (Fusion)

Section B: Werfen MATCH IT! Antibody

- B1: Create a Custom Export Template (One-Time Setup)
- B2: Export Your SAB Data
- B3: Working with Multiple Runs
- B4: Troubleshooting (MATCH IT!)

Part 2: Upload and analyze on HLAbAssist

- Accepted File Formats
- Convert Wizard: Preparing HAML Files
- Analyzing One SAB Panel for One Patient
- Analyzing Multiple SAB Panels for One Patient
- Multi-Patient Analysis
- Research Cohort Analysis

Part 1: Exporting Your SAB Data

Section A: One Lambda Fusion (LABScreen)

Reference: Based on the IHIW Lab07 guide (April 2026) and HLA Fusion v4.x.x User Manual (June 2024).

Werfen users: Skip to Section B.

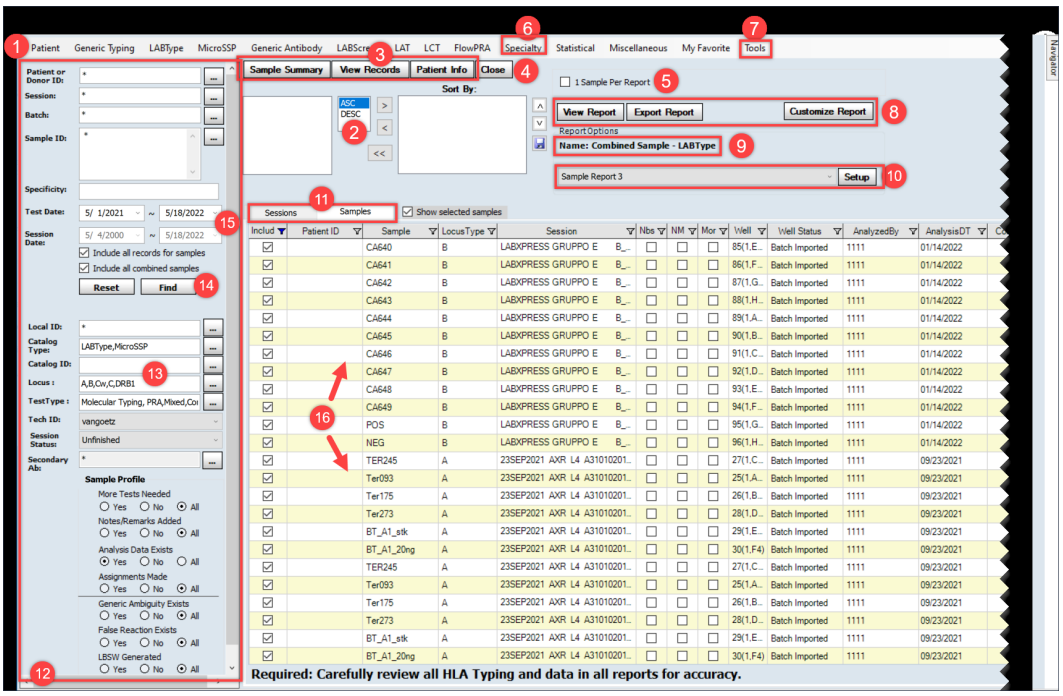
Before You Begin

Fusion must be installed with your LABScreen Single Antigen session already imported and analyzed.
Export one CSV per SAB panel (Class I and Class II separately).
Do not open the exported CSV in Excel. Excel silently reformats data (truncating leading zeros, converting bead IDs to dates) even without saving. Upload directly to HLAbAssist or the IHIW portal.

A1: Create the Export Template (One-Time Setup)

Step 1: Open the Reports Window

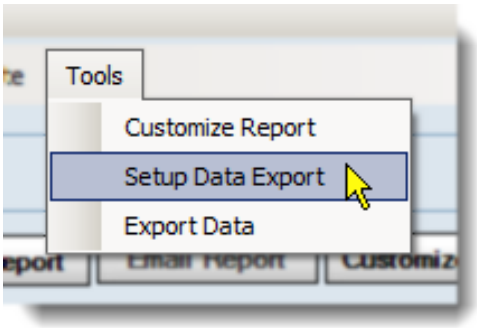
From the Fusion home page, click **Reports** on the toolbar or menu bar.



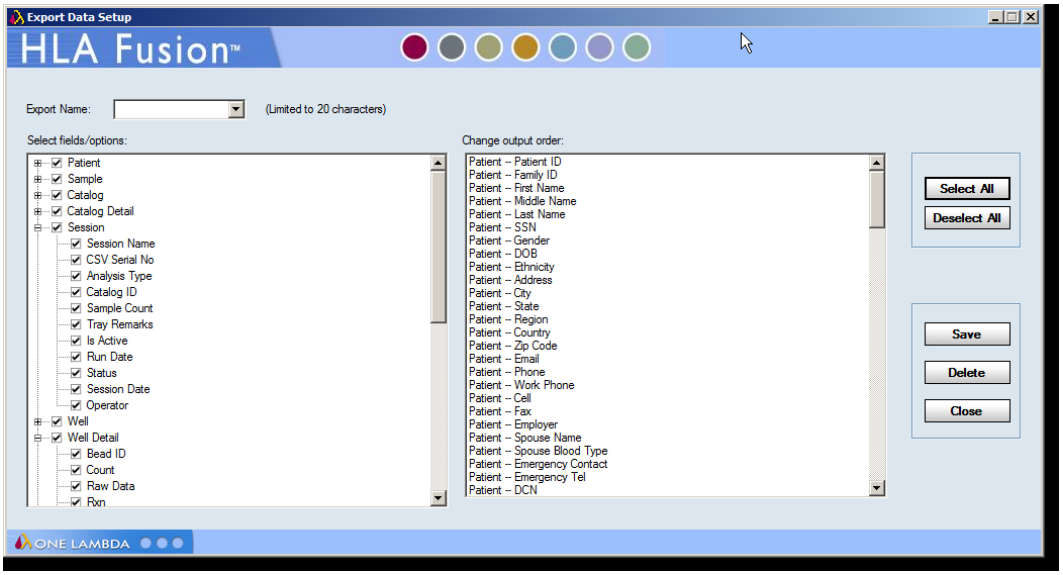
Fusion Reports window (Source: HLA Fusion v4.x.x User Manual): #1 Refine Report Input (Patient/Sample ID filters) · #7 Tools tab · #13 Catalog Type / Locus / Test Type filters · #14 Find button · #15 Session Date range · #16 Session list with Include checkboxes.

Step 2: Open Export Data Setup

Click the **Tools** tab (**#7**, top-right of the Reports window), then select **Setup Data Export**. The Export Data Setup dialog shows a name field at the top, a field tree on the left, and a field order list on the right.



Tools tab (#7) showing Setup Data Export option. Source: HLA Fusion v4.x.x User Manual.



Export Data Setup dialog: field tree (left) and output order (right). Source: HLA Fusion v4.x.x User Manual.

Step 3: Name the Template

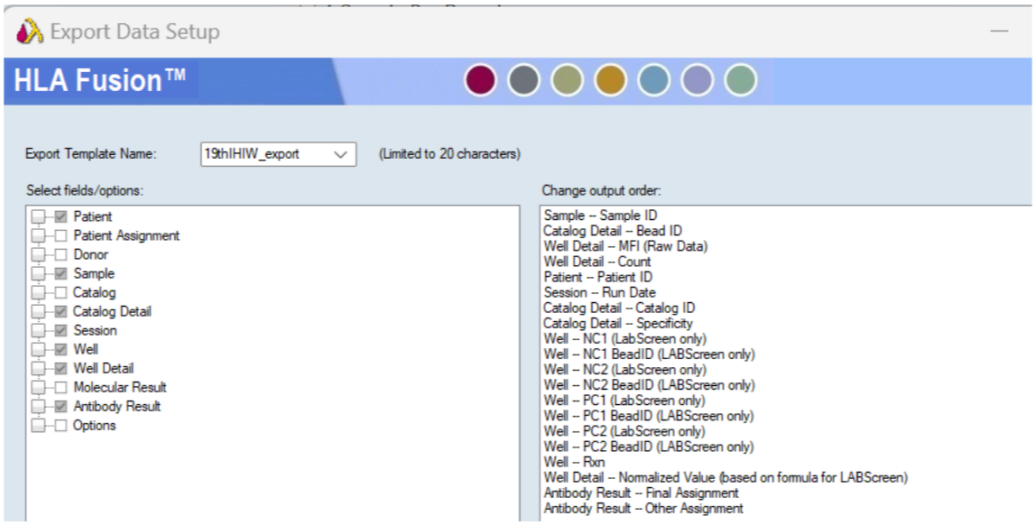
In the Export Template Name field, type what you wish to name the template.

Step 4: Select the Required Fields

Expand categories and check the boxes listed below. The right panel should show these 20 fields in this order:

#	Field	Category
1	Sample ID	Sample
2	Bead ID	Catalog Detail
3	MFI (Raw Data)	Well Detail
4	Count	Well Detail
5	Patient ID	Patient
6	Run Date	Session
7	Catalog ID	Catalog Detail
8	Specificity	Catalog Detail
9	NC1	Well
10	NC1 BeadID	Well
11	NC2	Well
12	NC2 BeadID	Well
13	PC1	Well
14	PC1 BeadID	Well
15	PC2	Well
16	PC2 BeadID	Well
17	Rxn	Well
18	Normalized Value	Well Detail
19	Final Assignment	Antibody Result
20	Other Assignment	Antibody Result

- Select the following elements:



Completed 19thIHIW_export template with all 20 fields selected. Source: IHIW Lab07 user guide (April 2026).

Step 5: Arrange and Save

Drag and drop (or use Ctrl + Up/Down) to match the order above. Click **Save**. This template is now available for all future exports.

A2: Export Your SAB Data

Repeat for each SAB panel (Class I, Class II).

Step 1: Open Reports

Click **Reports** from the Fusion home page.

Step 2: Select Your Session

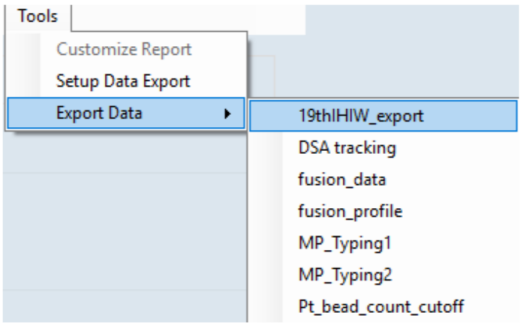
Set the **Session Date** range (**#15**) to include your SAB run.
Click **Find** (**#14**) to populate the session list (**#16**).
Locate your PROCESSED LABScreen Single Antigen session.
Check the **Include** box next to the session(s) to export.

> *Tip: Use the Catalog Type, Locus, and Test Type filters (#13) to narrow the list to LABScreen / Single Antigen only.*

Step 3: Export

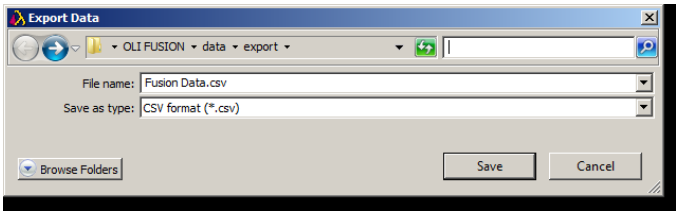
Click the **Tools** tab (**#7**).
Hover over Export Data. Select 19thIHIW_export from the submenu.

- Export CSV file:



Tools > Export Data submenu showing the 19thIHIW_export template. Source: IHIW Lab07 user guide (April 2026).

Navigate to a folder, enter a descriptive filename (e.g., ClassI_Serum01_20260422.csv).
In Save as type, select CSV format (*.csv). Click Save.



Export Data save dialog with CSV format selected. Source: HLA Fusion v4.x.x User Manual.

If prompted to open the file, click No.

A3: Exporting a Single Patient from a Multi-Patient Session

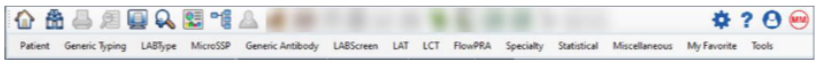
Fusion exports at the session level and may include multiple patients. Both HLAbAssist Single Patient mode and the IHIW portal expect one patient per CSV.

Option A: Filter in the Reports window (recommended)

In the Reports window, use the **Refine Report Input** panel (**#1**, left side).
Enter the patient's ID in the **Patient ID** field, or the sample ID in **Sample ID**.
Set the **Session Date** range (**#15**) and click **Find** (**#14**).

Select Report Type

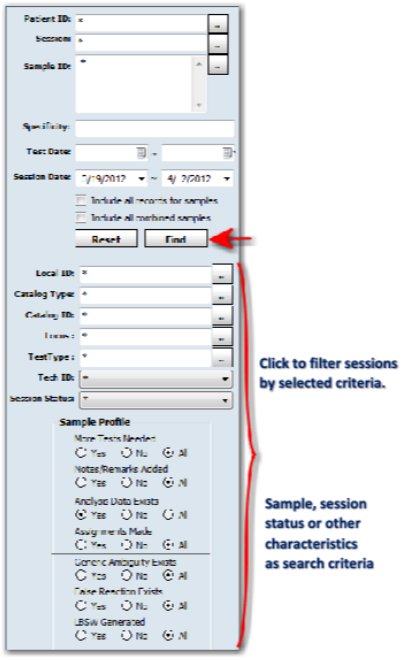
- Select a report from the report type menu options displayed at the top of the **Reports** window. The list of sessions in the right pane of the Reports window is filtered to display only the ones related to the selected report type.



Refine Report Input

If needed, use the left panel of the **Reports** window, to further filter the sessions you want to include in your report. There are a number of criteria you can set:

1. Enter a Patient ID, Session or Sample ID fields, or browse for the information with the **Browse** button.
2. Adjust the date range. Use the drop-down calendars in the **Session Date** fields to select a different start and end date.
3. Enter or browse for specific sample or session characteristics or status (see below).
4. Once you set criteria and click the **Find** button on the left panel of the **Reports** window, the session list in the right panel of the window filters accordingly.

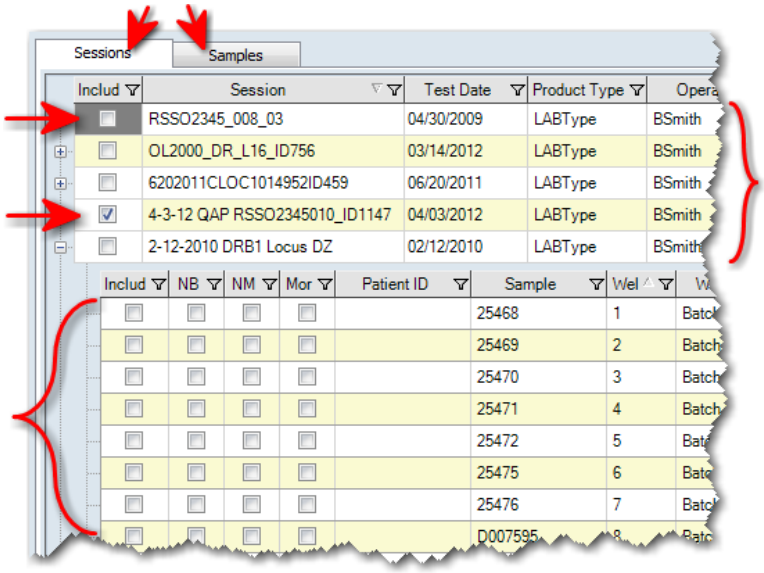


Session/Sample Selection

- In the Samples/Sessions list, click the + sign next to any session to expand the display to show its samples.

Refine Report Input panel (#1): enter Patient ID or Sample ID, set Session Date (#15), click Find (#14). Source: HLA Fusion v4.x.x User Manual.

Click the **+** sign next to a session in the list (**#16**) to expand it and see individual samples.



Session list (#16) expanded with + sign showing individual samples with per-sample Include checkboxes. Source: HLA Fusion v4.x.x User Manual.

Check **Include** next to only the sample you want; deselect others.
Export: Tools (#7) > Export Data > 19thIHIW_export > Save as CSV.

Option B: Split the CSV after export (do not use Excel)

Mac / Linux:

```
head -1 export.csv > single_patient.csv
grep "SAMPLE_ID_VALUE" export.csv >> single_patient.csv
```

Windows PowerShell:

```
$header = (Get-Content export.csv)[0]
$header | Out-File -Encoding utf8 single_patient.csv
Get-Content export.csv | Select-String "SAMPLE_ID_VALUE" | Out-File -Encoding utf8 -Append single_patient.csv
```

Replace SAMPLE_ID_VALUE with the actual Sample ID. Verify in a plain text editor: first row = column headers, every data row = same Sample ID.

A4: Troubleshooting (Fusion)

"I don't see the Tools tab" — Top-right of Reports window, next to "My Favorite." Older Fusion (pre-4.3): Reports menu > Tools > Setup Data Export.

"I don't see my session in the list" — Check Session Date range; confirm session Status is PROCESSED; ensure "Include all records for samples" is checked.

"My CSV looks wrong" — Verify in Notepad or TextEdit (not Excel): first row = column headers, remaining rows = comma-separated values.

"The portal or HLAbAssist rejects my file" — Confirm saved as CSV not XLS/XLSX; check it was not re-saved in Excel.

"My export contains multiple patients" — Normal for Fusion. See Section A3.

Section B: Werfen MATCH IT! Antibody (LIFECODES LSA)

Reference: Based on the IHIW Lab07 guide (April 2026), MATCH IT! Antibody v1.5 User's Manual (LC1684IVD Rev. D, April 2022), and LIFECODES LSA Product Insert (LC1683CEEN Rev. E).

Fusion users: Skip to Part 2.

Before You Begin

MATCH IT! Antibody v1.3 or later must be installed with your LIFECODES LSA session data imported and analyzed.

Export one CSV per SAB panel (LSA Class I and LSA Class II separately).

Confirm batch status: samples should be Completed or Approved before exporting. Completed samples appear bold; approved samples appear bold and italicized.

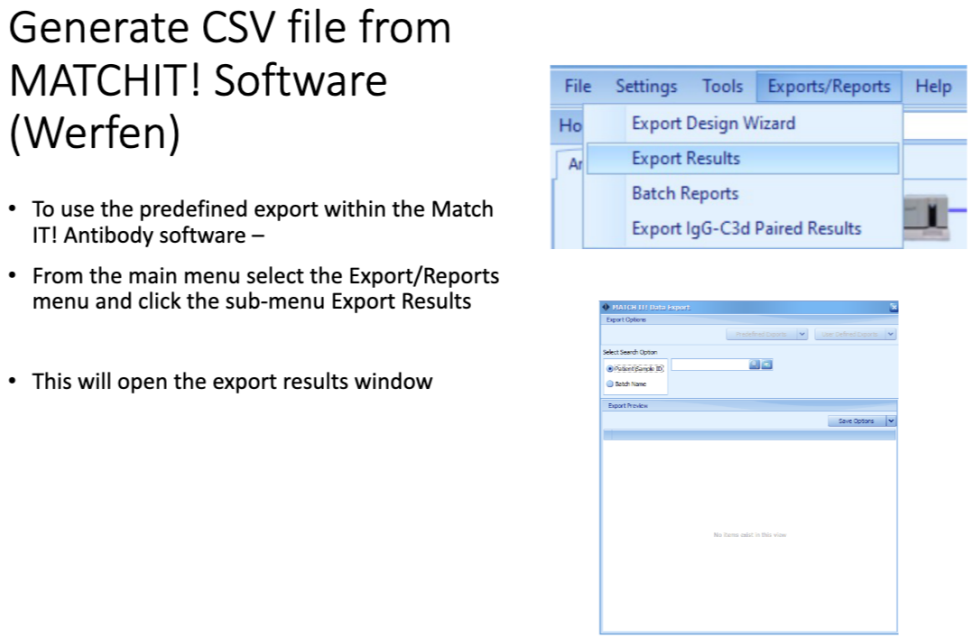
Do not open the exported CSV in Excel before uploading.

B1: Create a Custom Export Template (One-Time Setup)

MATCH IT! ships with two predefined exports — Antibody Tail Export and Antibody Tail ID — neither of which includes per-bead MFI data. You must create a custom template using the Export Design Wizard.

Step 1: Open the Export Design Wizard

From the MATCH IT! home screen, click **Exports/Reports** in the left menu, then click **Export Design Wizard**.



MATCH IT! Exports/Reports menu showing Export Design Wizard and Export Results options. Source: MATCH IT! Antibody v1.5 User's Manual.

Step 2: Select the Test Type

Select **LSA** (LIFECODES Single Antigen). One template works for both LSA Class I and LSA Class II.

Step 3: Select Export Columns

Check the boxes for the following fields:

#	Column Name	Description
1	Sample ID	Patient/sample identifier
2	Batch Name	Batch identifier assigned at import
3	Lot ID	LIFECODES bead lot number
4	Run Date	Luminex acquisition date
5	Bead ID	Luminex bead region number
6	Antigen	HLA specificity on the bead
7	Raw MFI	Unprocessed MFI
8	Count	Bead event count (low = unreliable)
9	MFI/LRA	MFI divided by Lowest Ranked Antigen MFI
10	BCR	Background Corrected Ratio
11	BG Adjusted	Raw MFI minus Negative Control bead MFI
12	Assignment	Positive / Negative / Weak call per bead
13	Serology	Serological equivalent of the allele
14	CON PC	Positive Control bead MFI
15	CON NC	Negative Control bead MFI

Optional but recommended: AD-MFI, AD-BCR, PRA.

Step 4: Set Delimiter and Name the Template

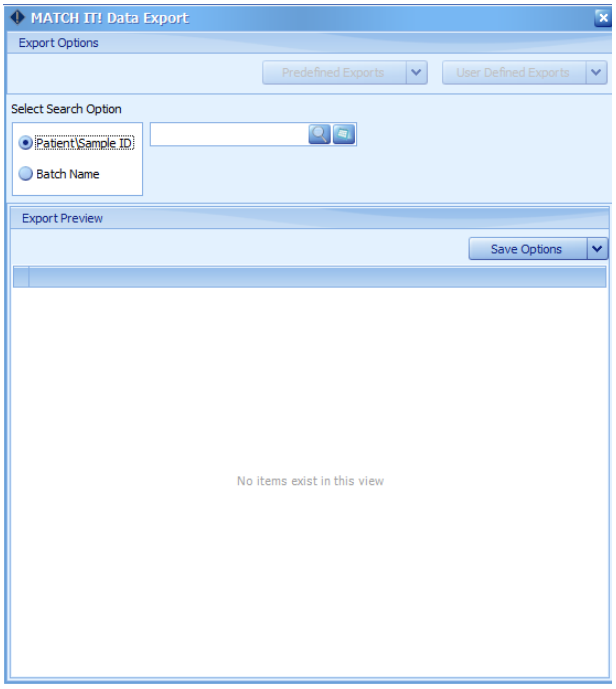
Select **Comma** as the field delimiter. Name the template: `19thIHIW_LSA`. Click **Finish** or **Save**.

B2: Export Your SAB Data

Repeat for each LSA panel.

Step 1: Open Export Results

From the MATCH IT! home screen, click **Exports/Reports** > **Export Results**.



Export Results window: search by Patient/Sample ID or Batch Name. Source: MATCH IT! Antibody v1.5 User's Manual.

Step 2: Search for Your Data

Choose a Search Option:

Patient / Sample ID: Type the ID and click the magnifying glass icon.
Batch Name: Type the batch name and click the magnifying glass icon.

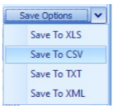
> *To export a single patient: always search by Patient / Sample ID. Searching by Batch Name exports all patients in the batch.*

Step 3: Select Template and Save

Click the User Defined Exports dropdown.
Select 19thIHIW_LSA.

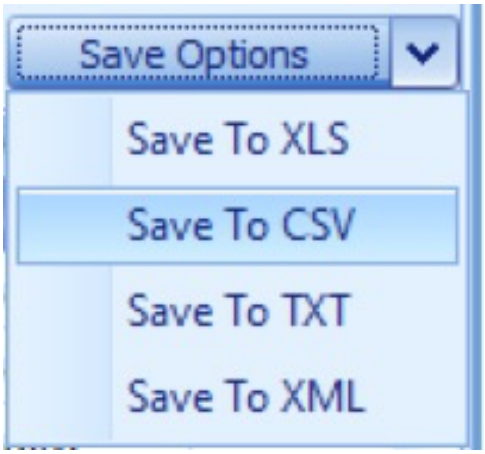
- Select your sample or batch that you would like to export. After selection the Predefined Exports dropdown will become enabled.

- Select the IHIW HAML export which will fill the grid below with the data and then choose the save option of CSV to save the data.



User Defined Exports dropdown showing 19thIHIW_LSA and Save Options dropdown. Source: MATCH IT! Antibody v1.5 User's Manual.

Click the dropdown next to Save Options > Save To CSV.



Save Options dropdown: select Save To CSV. Source: MATCH IT! Antibody v1.5 User's Manual.

Name the file descriptively (e.g., LSA_ClassI_Serum01_20260422.csv) and click Save.

B3: Working with Multiple Runs

For a patient with multiple SAB runs across different batches, export each batch separately. HLAbAssist accepts all CSV files dropped at once and merges them for temporal pattern analysis. The Panel (run date) selector will appear automatically.

B4: Troubleshooting (MATCH IT!)

"I don't see the Export Design Wizard" — Requires Technician or Lab Supervisor permissions. Ask your supervisor to create the template (Tools > Manage Users).

"I don't see my batch in search results" — Verify the batch was imported and analyzed. Try both search options (Sample ID and Batch Name).

"Some beads show FAILED" — Bead failed minimum event count. Double-click to override; CON bead failures must be resolved first.

"My CSV looks wrong" — Open in Notepad or TextEdit, not Excel.

"I use MATCH IT! older than v1.3" — Contact Werfen/Immucor support (855-466-8267 or www.immucor.com).

Appendix: Field Mapping Between Fusion and MATCH IT!

Concept	One Lambda Fusion	MATCH IT!
Sample identifier	Sample ID	Sample ID
Bead identifier	Bead ID	Bead ID
Raw signal	MFI (Raw Data)	Raw MFI
Bead event count	Count	Count
Bead lot	Catalog ID	Lot ID
HLA specificity	Specificity	Antigen
Run date	Run Date	Run Date
Negative control	NC1, NC2	CON NC
Positive control	PC1, PC2	CON PC
Normalized value	Normalized Value	MFI/LRA
Assignment	Final Assignment	Assignment
Background-corrected	(not standard)	BG Adjusted, BCR

Key difference: Fusion uses Normalized Value (NC-based); MATCH IT! uses MFI/LRA (LRA-based). Raw MFI is the most portable value for cross-platform comparison.

Part 2: Uploading and Analyzing on HLAbAssist

Accepted File Formats

HLAbAssist auto-detects your file format from the CSV column headers. No manual vendor selection is needed.

Format	Required Columns	How HLAbAssist Detects It
One Lambda Fusion CSV	Specificity, NormalValue (or Normalized), positive/negative control columns	Looks for Specificity + NormalValue headers
Werfen MATCH IT! CSV	BEAD_AG, BEAD_MFI_VALUE_ADJ, WS_ASSAY_PC, WS_ASSAY_NC	Looks for BEAD_AG or WS_ASSAY_PC headers
HistoTrac combined export	Mixed format (may contain both vendors)	Detects vendor per row from column patterns
HAML XML	Extended HAML format (<extended-haml> root)	Checks if file starts with <

Convert Wizard: Preparing HAML Files

The Convert wizard transforms SAB CSV into HAML (HLA Antibody Markup Language) XML for IHIW portal submission, archiving, or inter-lab sharing. PHI is stripped in the browser; patients receive pseudonymized IDs. Also available as Batch Convert CSVs → HAML on the Research cohort page.

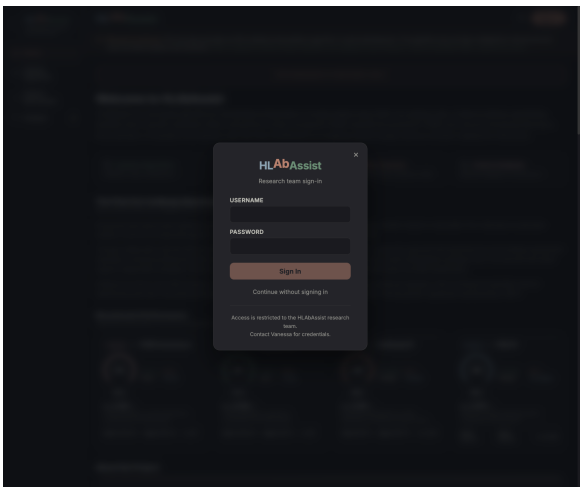
Privacy

HLAbAssist strips common PHI patterns in the browser before transmission. Only de-identified bead-level data and a pseudonymized patient ID are sent to the server.

Single Patient Analysis

Step 1: Open HLAbAssist

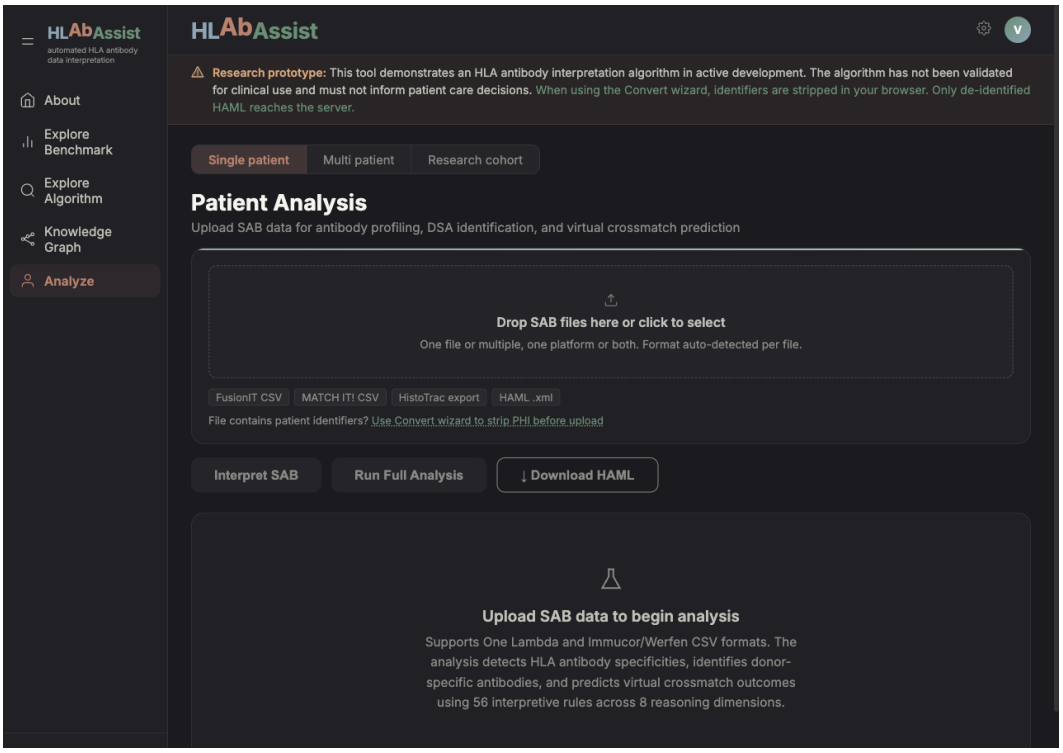
Go to **hlabassist.app** > **Sign In** > click **Analyze** in the left sidebar.



HLAbAssist sign-in dialog.

Step 2: Confirm Single Patient Mode

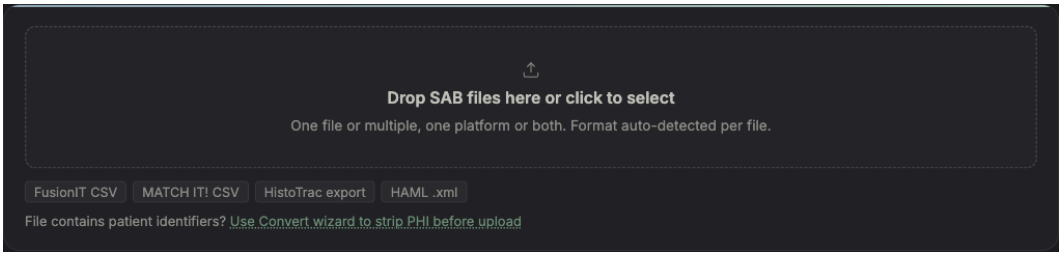
The Analyze page opens in **Single patient** mode by default.



HLAbAssist Analyze page in Single patient mode with the dashed upload drop zone.

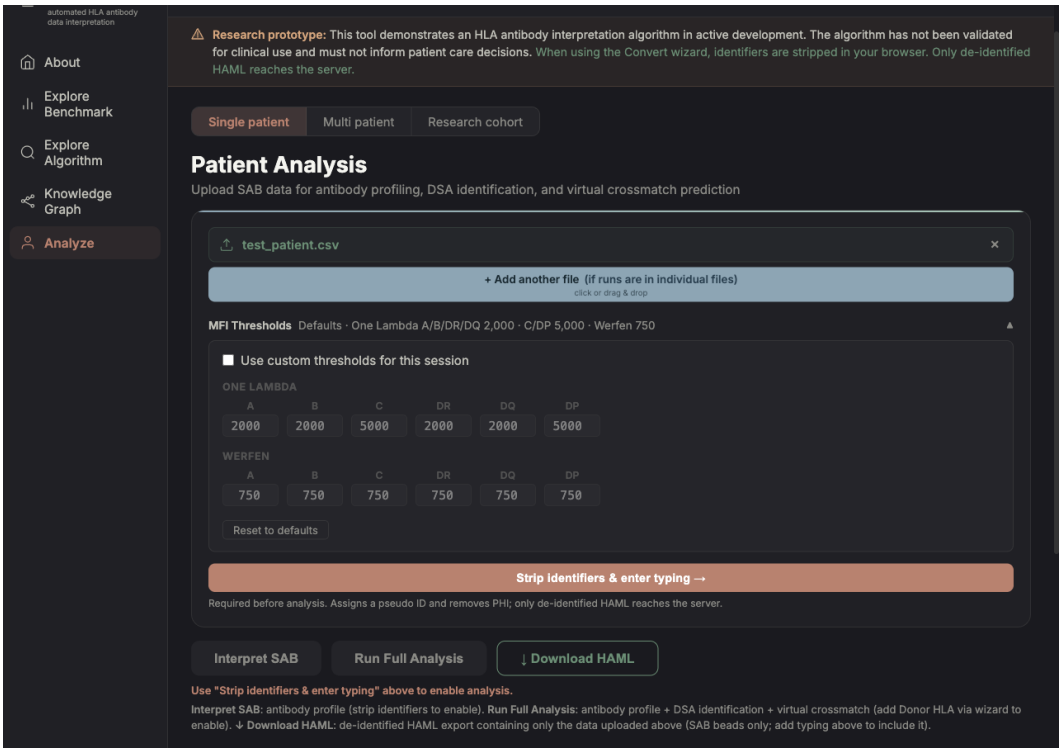
Step 3: Upload Your SAB File

Drag and drop your CSV onto the dashed drop zone, or click inside to browse. Accepted formats: FusionIT CSV (One Lambda), MATCH IT! CSV (Werfen), HistoTrac export (combined), HAML .xml (extended format).



Upload drop zone showing accepted format chips.

After uploading, default MFI thresholds are shown (One Lambda: A/B/DR/DQ 2,000 / C/DP 5,000; Werfen: all loci 750). These can be adjusted in the MFI Thresholds panel.

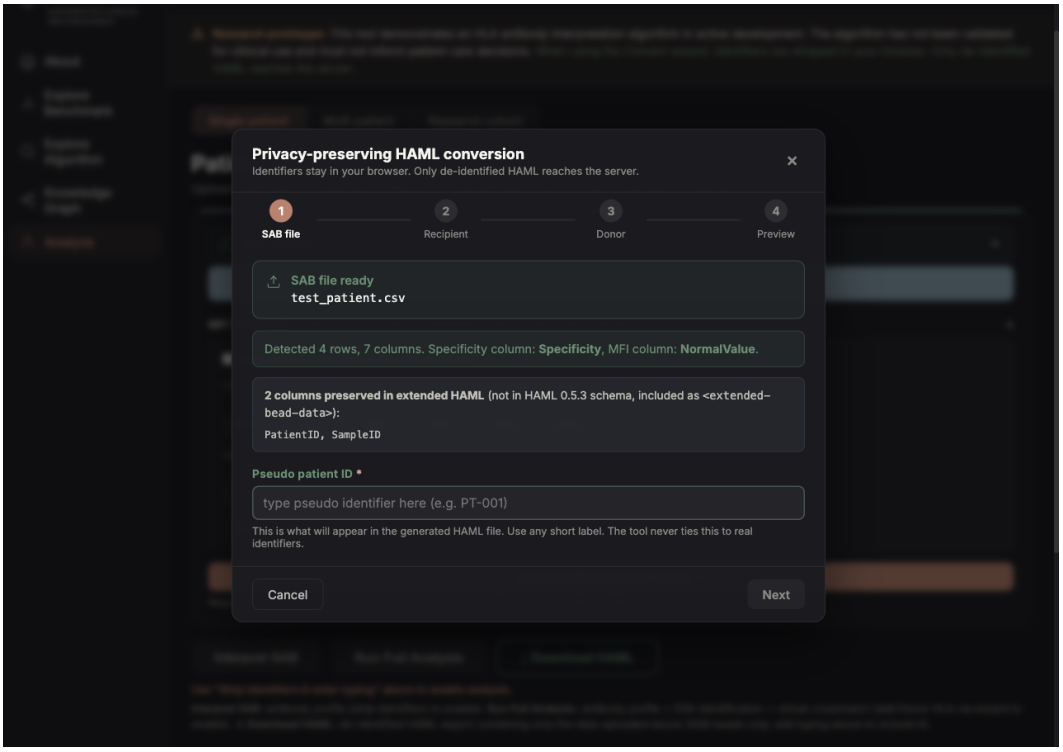


File loaded state: detected format chip, row count, and MFI threshold panel.

Step 4: Strip Identifiers and Enter Typing (Convert Wizard)

Click the orange Strip identifiers & enter typing → button. The wizard has four steps:

Step 1 — SAB file: Confirms row count and detected columns. Enter a Pseudo patient ID (e.g., PT-001).

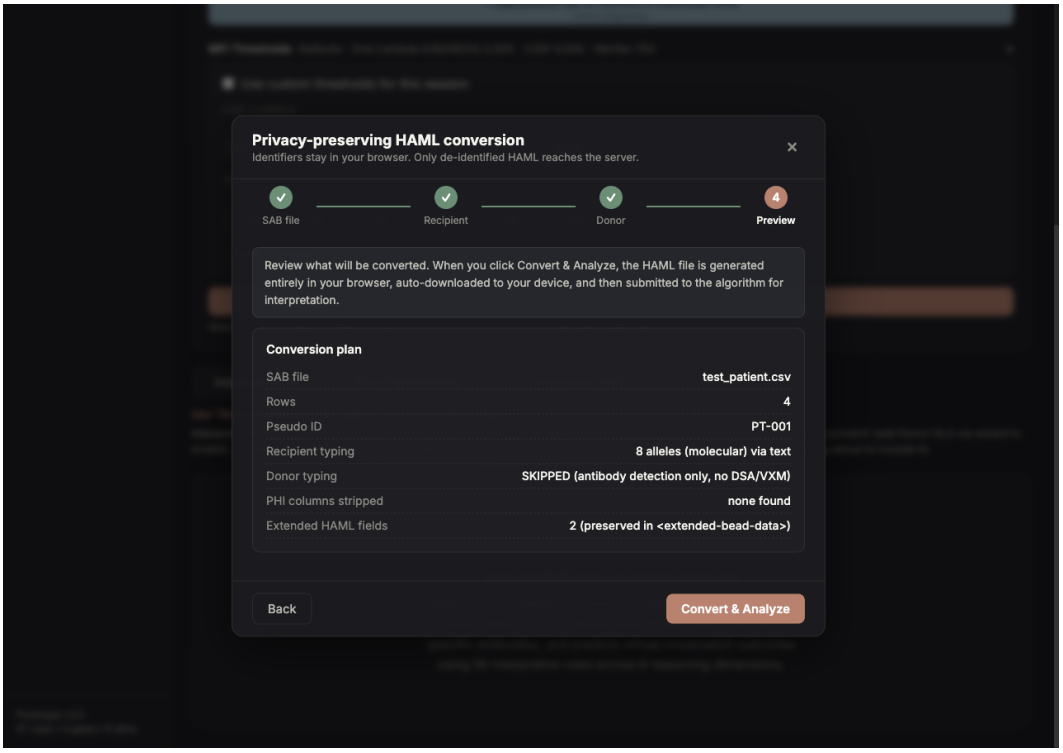


Convert wizard Step 1: SAB file confirmation and Pseudo patient ID entry.

Step 2 — Recipient: Paste recipient HLA alleles (e.g., A*02:01, B*07:02, DRB1*15:01) or upload a typing file. Required for DSA identification and self-antigen exclusion.

Step 3 — Donor: Paste donor alleles or upload a file. Required for VXM. Check Skip donor typing for antibody detection only.

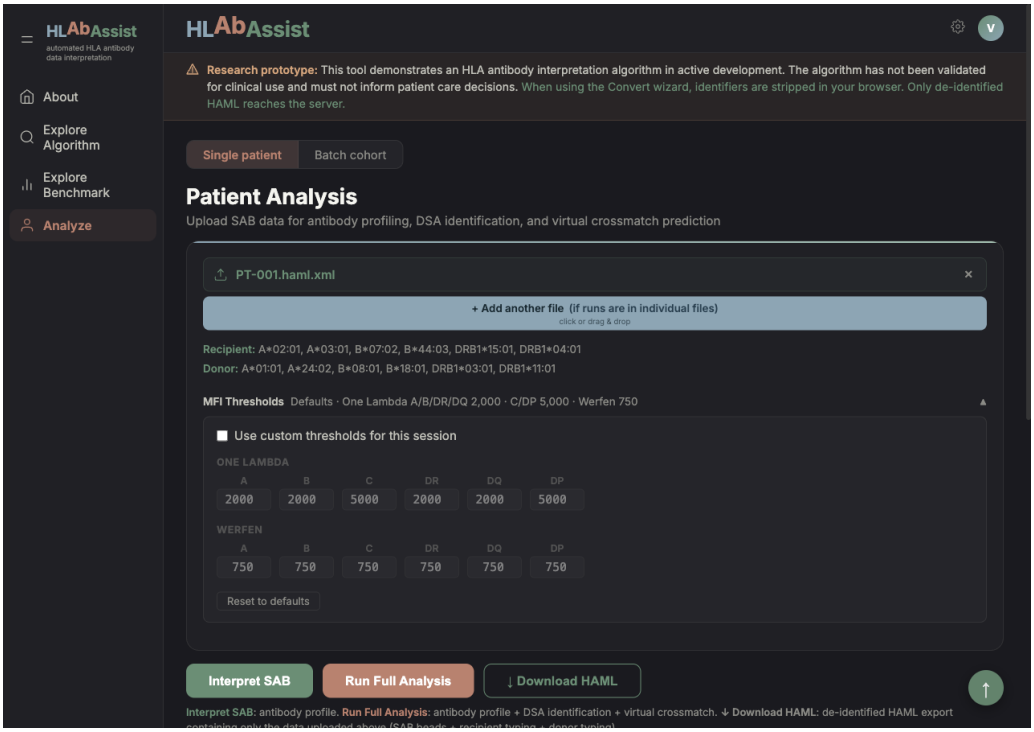
Step 4 — Preview: Review the conversion plan. Click Convert & Analyze. A HAML file is auto-downloaded. When the wizard closes, click Interpret SAB or Run Full Analysis to start the analysis.



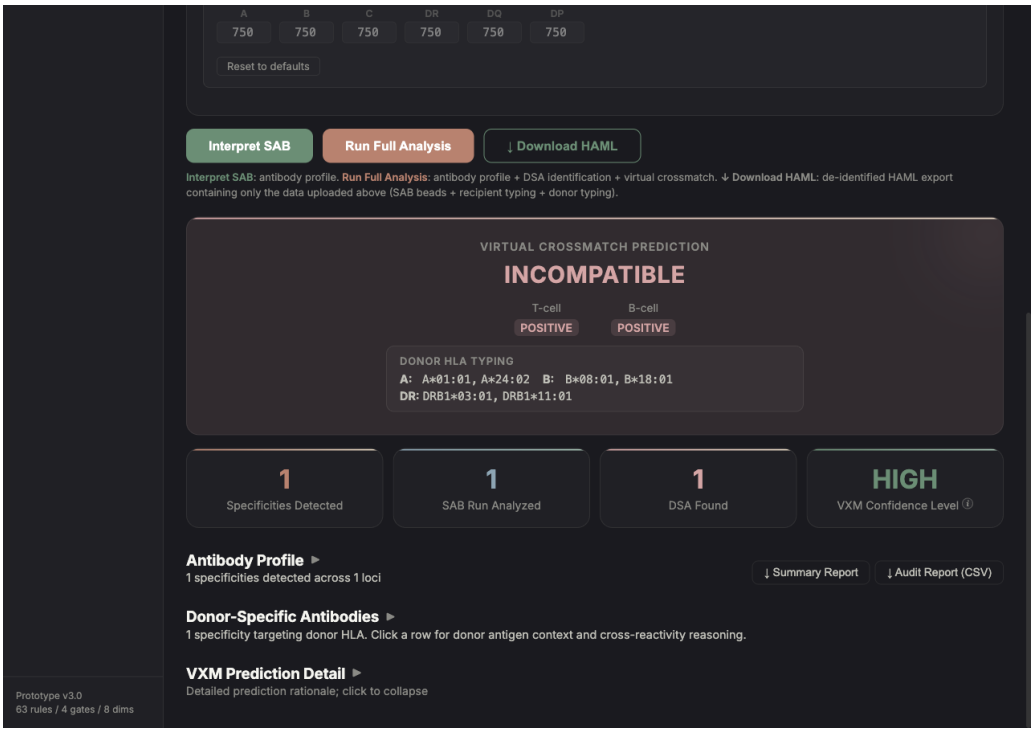
Convert wizard Step 4 (Preview): conversion plan summary before Convert & Analyze.

Step 5: Review Results

Virtual Crossmatch Prediction: INCOMPATIBLE or COMPATIBLE, with T-cell and B-cell predictions.
Summary cards: Specificities Detected, SAB Run(s) Analyzed, DSA Found, VXM Confidence Level.
Antibody Profile (expandable): Detected antibodies with MFI, confidence tiers, reasoning dimensions.
Download as Summary Report or Audit Report CSV.
Donor-Specific Antibodies (expandable): DSA with cross-reactivity reasoning.
VXM Prediction Detail (expandable): Rationale listing specific rules that fired.



Analysis results: VXM prediction card and summary statistics at the top of the page.



Analysis results scrolled: Antibody Profile and DSA expandable sections.

Analyzing Multiple SAB Panels for One Patient

Option A: Upload Multiple Files

Drop multiple CSVs onto the upload area. Common uses:

Class I + Class II panels (two Fusion CSVs or two MATCH IT! CSVs)
Two vendors for the same patient (one Fusion + one MATCH IT!)

Option B: Upload a Combined Multi-Run File

A CSV with a `RESULT_ID` column (common in HistoTrac or Werfen multi-batch exports) is automatically recognized as multiple runs.

The Panel Selector

When multiple panels are detected, a **Panel (run date)** dropdown appears:

All runs: cumulative profile (default) — analyzes all panels together; each antibody classified as persistent, waning, or new.
Individual panel — view a single timepoint only.

Multi Patient Analysis

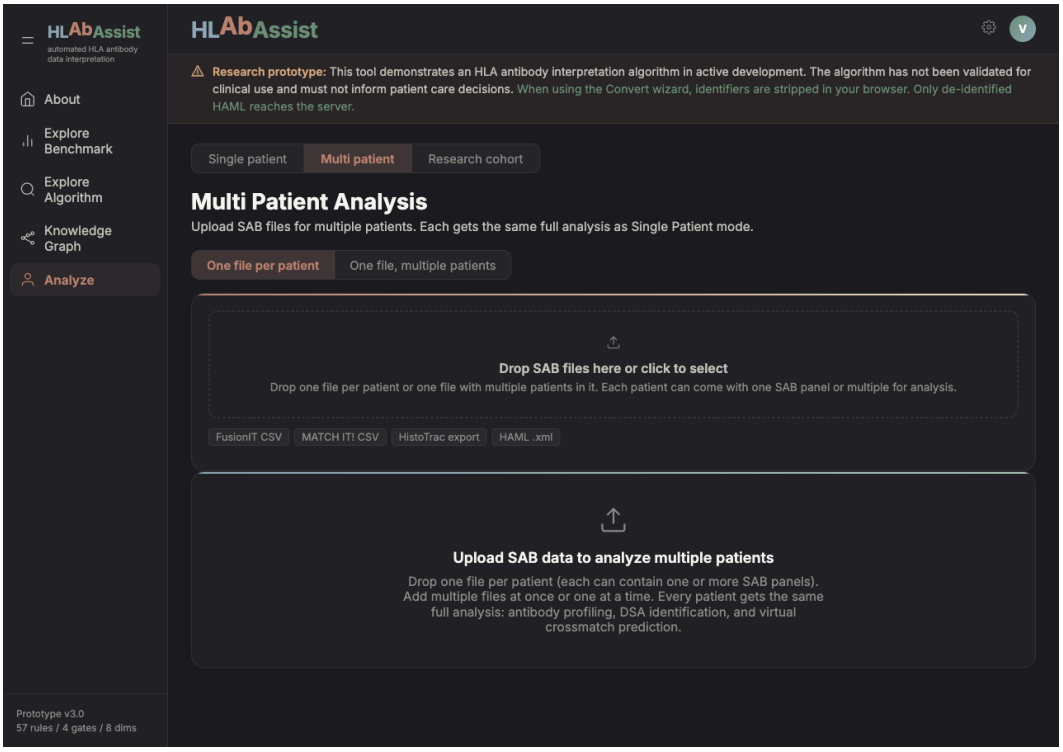
Use this mode for a caseload of several patients without population-level statistics. Two sub-tabs handle different file layouts.

Step 1: Switch to Multi patient Mode

Click the **Multi patient** mode button on the Analyze page (toggle at the top, between Single patient and Research cohort).

Sub-tab: One file per patient

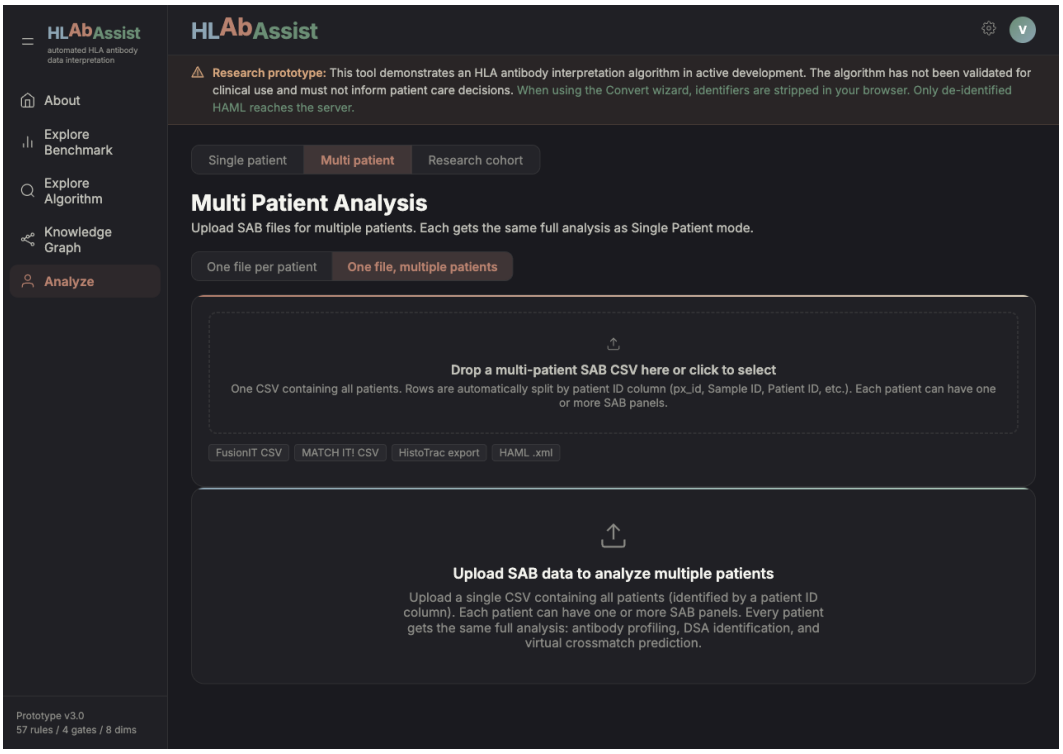
Use this sub-tab when each patient has their own separate CSV file. Drop one CSV per patient onto the upload zone — all files at once.



Multi patient — "One file per patient" sub-tab: drop zone for individual patient CSVs. Accepted formats: FusionIT CSV, MATCH IT! CSV, HistoTrac export, HAML .xml.

Sub-tab: One file, multiple patients

Use this sub-tab when your export is a single batch CSV containing rows for multiple patients. HLAbAssist splits the file automatically by patient ID column.



Multi patient — "One file, multiple patients" sub-tab: drop zone for a batch CSV that is auto-split by patient ID.

Step 2: Strip Identifiers and Convert

After uploading, click **Strip Identifiers & Convert**. HLAbAssist strips PHI per patient, assigns pseudonymized IDs (PT-001, PT-002, ...), generates one HAML per patient, and packages them in a ZIP archive (auto-downloaded).

Step 3: Review Results

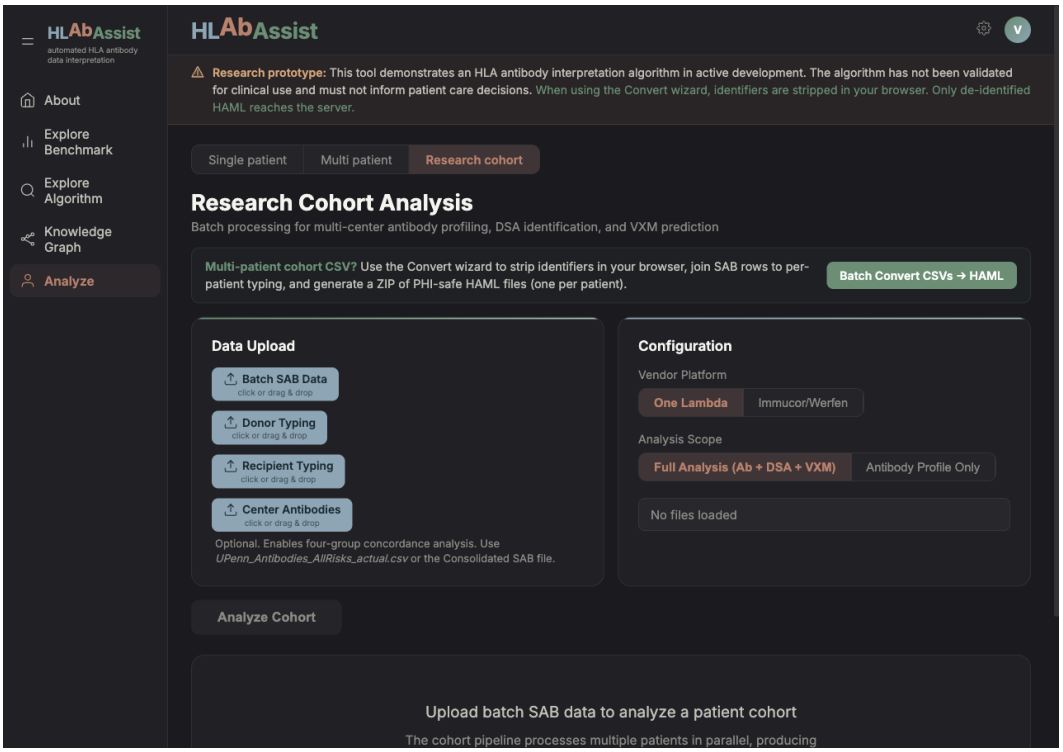
A summary table appears with one row per patient: Patient ID, Antibodies detected, DSA count, VXM prediction, Confidence level. Click any row to expand the full analysis.

Step 4: Export

Click **Download Summary CSV** or expand individual patients for Audit Report CSVs.

Research Cohort Analysis

Use this mode for population-level analysis.



Research cohort mode: upload slots for SAB data, donor/recipient typing, and center antibodies; vendor and scope selectors.

Step 1: Switch to Research Cohort Mode

Click the **Research cohort** mode button at the top of the Analyze page

Step 2: Upload Files

Upload Slot	Purpose	Required?
Batch SAB Data	Multi-patient SAB CSV with px_id column	Yes — Analyze Cohort disabled without it
Donor Typing	Donor HLA typing per patient	—
Recipient Typing	Recipient HLA typing per patient	—
Center Antibodies	Center-reported calls for concordance analysis	Optional (enables four-group concordance)

Step 3: Configure Analysis

Vendor platform: One Lambda or Immucor/Werfen (sets default MFI thresholds).
Analysis scope: Full Analysis (antibody + DSA + VXM) or Antibody Profile Only.

Step 4: Run and Review

Click **Analyze Cohort**. Results include a summary dashboard, distribution charts, four-group concordance (if center antibodies provided), DSA sensitivity table, and expandable per-patient rows.

Step 5: Export

Click **Export Results CSV** (summary) or **Export Detailed Report** (full per-patient detail).

Troubleshooting

"HLAbAssist doesn't recognize my file" — Check first-row headers in a plain text editor. Fusion needs Specificity + NormalValue/Normalized; MATCH IT! needs BEAD_AG or WS_ASSAY_PC. Re-export using Part 1 if headers differ.

"All antibodies are showing as one patient but I uploaded multiple" — You are in Single patient mode. Split your CSV (one per patient) or switch to Research cohort mode.

"I don't see the panel selector dropdown" — Requires a RESULT_ID column with more than one unique value, or two files uploaded. Single-panel patients do not show the selector.

"My MATCH IT! file has RESULT_ID but shows only one panel" — Confirm the CSV actually contains multiple unique RESULT_ID values.

"The analysis shows no DSA" — DSA requires recipient HLA typing in the Convert wizard Step 2. Without typing, only antibody detection runs.

"I see 'Werfen format detected' but expected One Lambda" — HistoTrac combined exports may include Werfen column names even for One Lambda beads; analysis accuracy is not affected.

Quick Reference Card

Single patient (1 panel):

hlabassist.app → sign in → Analyze → Single patient → drop CSV → orange button → pseudo ID + recipient + donor typing → Convert & Analyze

Single patient (multiple panels):

Analyze → Single patient → drop multiple CSVs → Convert wizard → use "Panel (run date)" dropdown for individual panels or cumulative profile

Multi patient (individual files):

Analyze → Multi patient → drop one CSV per patient → (optional) typing CSV with px_id → Strip Identifiers & Convert → click row to expand

Multi patient (batch CSV):

Analyze → Multi patient → drop CSV with px_id column → (optional) typing CSV → Strip Identifiers & Convert

Research cohort:

Analyze → Research cohort → upload SAB CSV (px_id required) → (optional) donor/recipient/center files → select vendor + scope → Download CSV

For questions about HLAbAssist, visit hlabassist.app or contact the development team at vmenard@tulane.edu.

For Fusion software support: techsupport@onelambda.com or 747-494-1000.

For MATCH IT! software support: Werfen/Immucor Technical Support (855) 466-8267 or www.immucor.com.